THz Oscillation in Doped-GaN Based Planar Gunn Diode with the T-shape Channel

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***Abstract* — In this study, the performance of a doped-GaN-based planar Gunn diode (PGD) with a T-shape channel is investigated through numerical simulations. The effects of the vertical electric field and the surface geometry are considered simultaneously through three-dimensional (3D) simulations. For comparison, 2D simulations are performed on the top view of the 3D structure. The dependence of the velocity on the electric field and doping density is calibrated. Differences in I-V characteristics and oscillation behaviour between 2D and 3D simulations are demonstrated. A high-frequency mode has been observed in 3D simulations at high voltage due to the formation of multiple dipole domains, which are not well represented in 2D simulations. The critical voltage of the low-frequency mode to high-frequency mode transition is found to reduce with the active layer thickness. With a 150 nm thick active layer, the oscillation frequency at 30 V estimated by the 3D simulation is higher than 600 GHz.**

***Keywords* — GaN, planar Gunn diode, Terahertz, TCAD simulations.**

I. Introduction

Terahertz (THz) range has attracted widespread attention due to its broad application prospects in diverse fields, including but not limited to chemical analysis, biology-imaging, and communication [1]. Significant efforts have been made to develop THz sources using optical and electrical technologies. Among them, the Gunn diode, which exploits the negative differential mobility (NDM) of III-V materials, is an attractive option due to its simplicity and low noise [2]. Compared to GaAs and InP, GaN provides a larger bandgap and higher electron velocity, making GaN Gunn diode promising for THz oscillations. Recently, GaN planar Gunn diode (PGD) has been extensively studied [3–5]. Unlike the vertical Gunn diode with the $n^+/n/n^+$ structure, whose frequency is determined mainly by layer thickness and fixed after the epilayer growth, the oscillation frequency of the PGD can be controlled by the channel length. In addition, various surface geometries can be integrated with the planar structure to enhance the local electric field and promote the onset of the Gunn oscillations [4, 5].

The shaped-channel PGDs based on AlGaN/GaN HEMT heterostructures, which rely on two-dimensional electron gas (2DEG), were proposed in [3, 6]. The 2nd harmonic frequency of 338 GHz has been predicted for the structure with a grooved anode using 2D simulations. The effect of surface geometries such as T-shape and V-shape has also been investigated using 2D simulations of the top view of the structure [5]. It has been reported that the oscillation frequency of a 500 nm long device can reach 450 GHz [5]. Recently, doped-GaN-based PGDs have been proposed as an alternative, which can further reduce the threshold voltage (V_{th}) and provide more frequency freedom by changing the doping concentration, and the optimal epilayer structure has been suggested [7]. Combined with surface geometries, the V_{th} can be reduced further by focusing the field near the cathode. However, the utilisation of doped-GaN and shaped channels have yet to be explored. The vertical electric field at the interface of doped-GaN and GaN buffer may also play an essential role in modulating Gunn oscillations, which can only be evaluated concurrently with surface geometries in three-dimensional (3D) structures. Hence, it is necessary to explore the effect of the shaped channel through 3D simulations compared to 2D simulations to accurately evaluate the oscillation performance of doped-GaN-based PGD with surface geometries. However, there is a lack of knowledge regarding this kind of study.

This work presents a detailed study of the oscillation performances of the doped-GaN-based PGD with a T-shape channel based on the TCAD simulator. The difference in oscillation performances evaluated by 2D and 3D simulations is investigated. Additionally, the effect of the active layer thickness on Gunn oscillations is discussed.

II. Simulation Model and Calibration

The schematic structure of the doped-GaN-based PGD studied in this work is shown in Fig. 1(a). 3D simulations were performed on this structure using TCAD Silvaco Atlas. It consists of an active GaN layer with a doping concentration of 2×10^{18} cm^{-3} and a 1 μm thick UID (unintentionally doped) GaN buffer layer with a carrier concentration lower than 5×10^{17} cm^{-3}. The thickness of the active layer t_{ac} varies from 300 nm to 150 nm in the simulations. In practice, the epistructures of doped-GaN-based PGD with various t_{ac} are grown by PA-MBE. The T-shape channel is formed by etching the vertical and horizontal trenches, defined as air in the

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simulation. The trench depth D_{trench} is set to 300 nm to reduce the leakage current. The width of the vertical and horizontal trenches, as indicated by W_{vt} and W_{ht}, is set to 200 nm and 100 nm, respectively. The channel length L and width W are set to 500 nm and 150 nm, respectively. To reduce the computation time and maintain accuracy, the distances from the channel to the cathode and anode are set as short as possible to 100 nm and 600 nm, respectively, since it does not impact the oscillations [5]. For comparison, 2D simulations are conducted on the top-view structure, as illustrated by Fig. 1(b). To evaluate the current level with various t_{ac} in 2D simulations, the thickness along the missing axis, which is Y-axis in Fig. 1(a), was set to be consistent with t_{ac}. The self-heating effect is not considered in this work and the lattice temperature is maintained at 300 K. To understand the difference in the oscillation behaviour between 3D and 2D simulations, the electric field distributions along the X-axis at the centre of the channel (Z = 0.75 µm/2) in the 2D structures is compared with that found in the middle of the doped layer (Y = t_{ac}/2) in 3D structures.

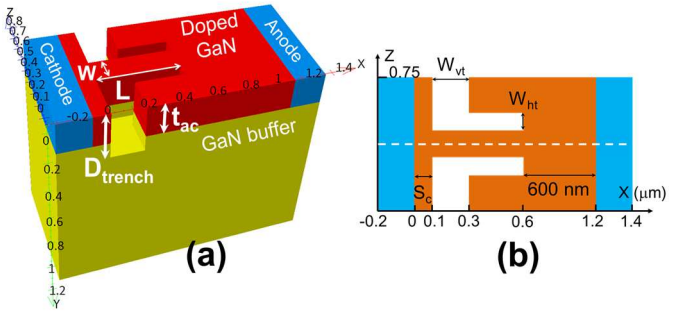


Fig. 1. 3D structure of the doped-GaN based PGD with T-shape channel (a) and the top view structure used in the 2D simulations (b).

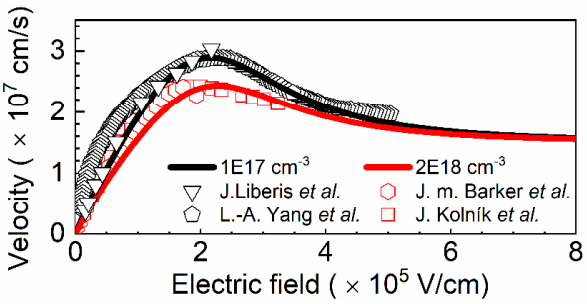


Fig. 2. Velocity as a function of the electric field.

In TCAD simulations, the migration of electrons from the lower valley to the upper valleys under a high electric field resulting in the increased effective mass and reduced velocity is accounted for by the NDM model as expressed by:

$$u = \frac{u_0 + v_{sat} * \frac{E^{\gamma-1}}{E_c^{\gamma}}}{1 + \alpha * \left(\frac{E}{E_c}\right)^{\beta} + \left(\frac{E}{E_c}\right)^{\gamma}} \quad (1)$$

where u_0 represents the low field mobility, calibrated according to the Hall measurements on the epilayers grown by PA-MBE and *I-V* characteristics of a 6 µm long device without the T-shape channel [8, 9]. Furthermore, the dependence on the electric field has been calibrated according to the reported Monte Carlo (MC) data and experimental data of Si-doped GaN at the concentration of 10^{17} [10, 11] and 2×10^{18} cm^{-2} [12, 13], as shown in Fig. 2. The fitting parameters are listed in Table 1.

Table 1. Fitting parameters for (1).

v_{sat} (cm/s)	E_c (V/cm)	α	β	γ
1.50×10^7	2.35×10^5	0.57	0.80	4.4

III. Results and Discussion

A. *I-V* characteristics

Transient simulations were performed on 2D and 3D structures at different anode voltages with a total simulation time of 10 ps. The time-step was set to 0.05 ps, which is much smaller than the energy relaxation time τ_ε of 0.5 ps [3, 14, 15]. To ensure the simulation accuracy while limiting the calculation time, the mesh size along the X-axis was set to 5 nm which works well even though it is slightly higher than the Debye length which is ~3 nm for GaN doped at 2×10^{18} cm^{-3}.

The *I-V* characteristics shown in Fig. 3(a) were obtained by calculating the average current at each voltage. The corresponding current oscillations are shown in Fig. 3(b). As expected, the current level is proportional to the t_{ac} in both 2D and 3D simulations. The *I-V* characteristics of 2D simulations demonstrate a linear increase region followed by a saturation region when the voltage is higher than ~15 V. In addition, the current reduction at ~15 V due to the NDM is clearly observed in 2D simulations, consistent with the dipole domain formation in the channel and the onset of current oscillations.

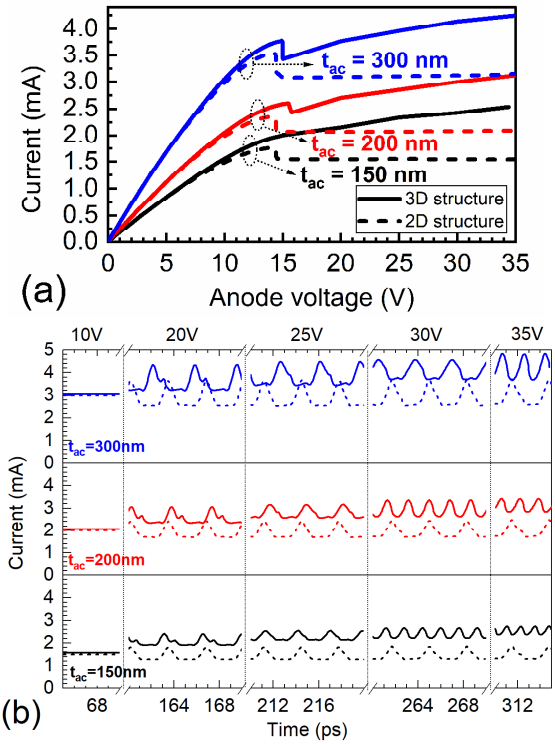


Fig. 3. *I-V* characteristics of 2D structures (dashed lines) and 3D structures (solid lines) with various t_{ac} (a) and the corresponding current oscillations at different voltages (b).

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By including the UID GaN buffer layer in 3D simulations, the same current level as in 2D simulations is observed in the linear region, indicating low leakage current in the buffer layer at low bias. However, as the applied voltage increases, a more significant difference in the current levels between 3D and 2D simulations has been observed. This indicates higher leakage current flow through the buffer layer under high voltages, which may degrade the power conversion efficiency. Hence, it is suggested by this work to enhance the electron confinement at the interface of doped-GaN/UID GaN, for example, by inserting an AlGaN back-barrier layer. Similar V_{th} to the 2D simulations has been obtained in 3D simulations, as shown in Fig. 3(a) where the current abruptly drops. Notably, there is no apparent current reduction as t_{ac} is reduced below 200 nm. This can be attributed to the strong contribution of the leakage current as compared to the low oscillation amplitude with thinner active layers. Nevertheless, stable current oscillations have been still observed at voltages above ~16 V for the structure with a t_{ac} of 150 nm.

B. Oscillation performance

In addition to *I-V* characteristics, the oscillation performances in 2D and 3D simulations are also compared. Fast Fourier Transform (FFT) has been performed on the time-domain current oscillations. Fig. 4(a) and (b) show the frequency and amplitude of the fundamental oscillation as a function of the applied voltage. With a t_{ac} of 300 nm, the frequency in 2D simulations remains almost unchanged at ~300 GHz. Fig. 5(a) shows the electric field profiles along the X-axis in the 2D structure at 35 V. The inset shows the corresponding current oscillation in the time domain. From t_1 to t_3, the 1st dipole domain reached the channel exit near the anode and gradually disappeared. This leads to the gradual increase of both the current and the electric field at the entrance of the channel near the cathode, and a new domain starts to be formed (labelled as 2nd). Note that the 1st domain only partially disappeared at t_3. While the fading away of 1st domain tends to increase the current, the decrease in current due to the rapid growth of 2nd domain dominates the evolution from t_3 to t_5. With the 2nd domain matured at t_5, the current reaches the minimum and is maintained during the migration towards the channel exit near the anode. As a result, the current oscillates at a frequency of about 300 GHz.

Similar oscillation frequencies are observed for the 3D structure when the applied voltage is below 30 V. However, further increasing of the voltage to 35 V leads to an increase in frequency from ~300 GHz to above 500 GHz. To explain this, the electric field profiles in the middle of the channel at the plane of Y = 0.1 µm are shown in Fig. 5(b). Unlike the 2D simulations, it is found that two dipole domains exist simultaneously in the channel from t_1 to t_3. The disappearance of the 1st domain that dominates the current change leads to both current increase and the formation of the 3rd dipole domain. The trade-off between the current changes caused by the vanishing and growth of multiple dipole domains enhanced harmonic components and is well-captured in 3D simulations. Therefore, much higher oscillation frequencies are obtained. The operating mode of the doped-GaN PGD under such conditions is defined as the 'high-frequency mode' in this work. This indicates that simulations of 2D structures might not be able to accurately capture the formation of multi-domains in the T-shape channel of the doped-GaN-based PGD, especially under high applied voltages. Hence, it is reasonable to suggest 3D simulations, especially for high applied voltages.

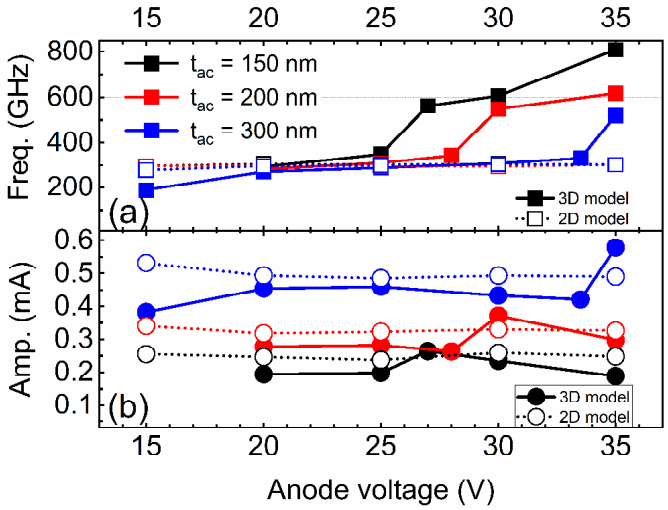


Fig. 4. Fundamental frequency (a) and amplitude (b) as a function of anode voltage obtained from 2D (dotted lines with open symbols) and 3D (solid lines with closed symbols) simulations, respectively.

C. The effect of the active layer thickness

The effect of the value of t_{ac} on the oscillation performances has been investigated by varying it from 300 nm to 150 nm in the 3D structure. As shown in Fig. 4, low-frequency and high-frequency modes have been observed for all t_{ac} in 3D simulations. It is shown that the high frequency mode is reached for lower voltage when t_{ac} is reducing, providing oscillations with frequencies above 600 GHz. The electric field profiles shown in Fig. 6(b) and (c) confirm the formation of the three dipole domains that induce high-frequency oscillations for the thinner t_{ac} of 200 nm and 150 nm. In comparison, only two domains exist simultaneously for t_{ac} of 300 nm, as shown in Fig. 6(a). Hence, using thinner active layers reduces the voltage for the onset of multiple dipole domains and thus enters the high-frequency mode.

Nevertheless, similar frequency and amplitude trends have been observed as the voltage is increased for all t_{ac} studied in this work. In the low-frequency mode, the oscillation obtained in 3D simulations shows a similar frequency as 2D simulations, whereas the amplitude is slightly lower. As the voltage is increased, a transition from low-frequency to high-frequency mode occurs with an increase in the oscillation amplitude. Further increasing of the anode voltage leads to higher oscillation frequencies at the expense of the amplitude.

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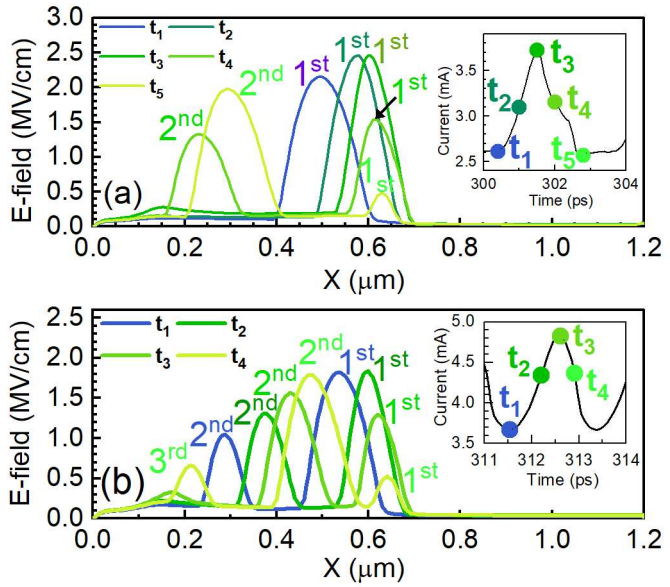


Fig. 5. Electric field profiles at the middle of the channel along the X-axis at V_{dc} of 35 V in 2D (a) and 3D (b) structures with a t_{ac} of 300 nm. The insets show the corresponding current oscillations of long-time sequences and the times selected for the representation of the electric field profiles.

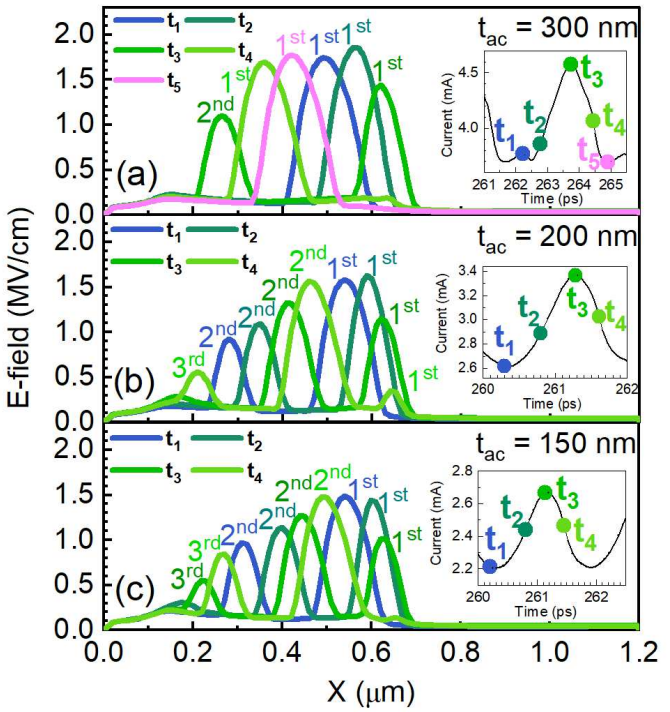


Fig. 6. Electric field profiles at the middle of the channel along the X-axis at 30 V in 3D simulations with t_{ac} of 300 nm (a), 200 nm (b), and 150 nm (c). The insets show the corresponding current oscillations of long-time sequences and the time selected for the representation of the electric field profiles.

IV. Conclusion

The THz oscillations in the doped-GaN-based PGD with the T-shape channel have been studied through 3D numerical simulations considering both the effect of the vertical electric field and the T-shape channel. For comparison, 2D simulations have also been conducted on the top view of the structure. It is found that 3D simulations exhibit higher current levels due to the leakage current in the GaN buffer layer when the anode voltage is above the V_{th}. The oscillation behaviour is also different in 2D and 3D simulations. The high-frequency mode with the formation of multiple dipole domains, which is not accurately captured in 2D simulations, has been observed in 3D simulations. 2D and 3D simulations show a similar oscillation frequency of ~ 300 GHz at low-frequency mode, almost independent of voltage. In the high-frequency mode, the frequency has been found to increase as the voltage with the reduced amplitude. The critical voltage for the transition from low-frequency to high-frequency mode has been found to decrease as the t_{ac}. Oscillation with a frequency higher than 600 GHz has been observed at 30 V with a t_{ac} of 150 nm.

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